Proposal:

1. **What problem did you select and why did you select it?**

   We chose to automate and optimize professional resume creation, addressing two key challenges: (a) the tedious, manual process of tailoring a resume to each job posting, and (b) the rigid filters used by Applicant Tracking Systems that often screen out qualified candidates. By building a system that parses job descriptions and refines resume content automatically, we help users save time and improve their success rate with both human recruiters and ATS.

---

2. What database/dataset will you use?

- **Primary data:** The user's own uploaded resumes and corresponding job descriptions (our "live" data).

---

3. What NLP methods will you pick from the concept list? Will it be a classical model or will you have to customize it?

- **Transformer-based extraction:** spaCy's transformer NER for entities (company, location), RoBERTa QA for responsibilities/qualifications.
- **Semantic matching:** Sentence-BERT embeddings for keyword and skills alignment.
- **Prompt engineering & LLM generation:** MistralAI (via Together/OpenAI) to rewrite and format sections per ATS rules.
  We do **not** train classical models from scratch—our "customization" consists of crafting and tuning prompts rather than developing new architectures.

---

4. What packages are you planning to use? Why?

- **spaCy**: Industry-standard NER with transformer support.
- **transformers** & **openai**: For QA pipelines and ChatCompletion calls to GPT-4/MistralAI.
- **sentence-transformers**: Easy, high-quality semantic embeddings (Sentence-BERT).
- **scikit-learn**: TF-IDF vectorization and evaluation metrics.
- **Streamlit**: Rapid development of an interactive web UI.
- **subprocess** / **PyLaTeX**: Automate LaTeX compilation of the final PDF.
  These libraries are well-documented, GPU-optimized, and widely used in both research and production.

---

5. What NLP tasks will you work on?

1. **Named Entity Recognition and REGEX** (Extraction of Details)
2. **Question Answering** (responsibilities, qualifications)
3. **Embedding-based Retrieval** (skill matching via cosine similarity)
4. **Text Generation** (LLM-driven rewriting of resume sections)
5. **Keyword Weighting** (TF-IDF to prioritize terms)

---

6. How will you judge the performance of the model? What metrics will you use?

- **ATS Score: It gives us an Idea of how the resume uploaded is good enough for the Description**
- **If any models like BERT or transformers are used we will mention the model score**

---

7. Provide a rough schedule for completing the project.

Week	Focus	Deliverables
1.	Extraction	Job Description and Resume Extraction
2	UI	Creating Interactive UI
3	Prompt engineering	Draft prompts for each resume section
4	Backend integration	Orchestrator linking parser → LLM → cleanup

These is the Draft, throughout the weeks the particulars may change but the end goal remains same..